

Dynamic Analysis Establish Settings Procedure

Revision

Version 1
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SME

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Abstract

This document describes the procedure used to perform the threat model creation activity described in the **AVCDL** ^[1] secondary document **Dynamic Analysis Report** ^[2].

Group / Owner

Security / Security Architect

Motivation

This document is motivated by the need to have runtime-specific, cybersecurity-related feedback in the development of software for use within safety-critical, cyber-physical systems.

Note: Within the context of this document, the terms ***security*** and ***cybersecurity*** are used interchangeably. It is presumed that the term ***security*** is being used in reference to ***cybersecurity*** and not ***physical security***.

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Audience

The audience of this document is the cybersecurity practitioner who will be conducting cybersecurity-specific dynamic analysis.

Completeness of Output

Since dynamic analysis is an as-is analysis, it is critical to ensure that the information gathered is as accurate and complete as possible. As with any cybersecurity assessment, it is not the place of the cybersecurity SME to make assumptions on the part of engineering. All information should be sourced from and confirmed by the owner of the element under consideration.

When information is not available for either a given section of the template or parts thereof, this should be noted. Major omissions should be recorded in the cybersecurity risk register.

Disposition of Output

Once completed, the generated output should be managed in the organization's document management system (DMS) as a document of record.

Entry Criteria

This document assumes that the reader understands the purpose of cybersecurity threat modeling. Further, that the reader has read and understood the AVCDL **Dynamic Analysis Report** [\[2\]](#) and **Secure Settings Document** [\[3\]](#) secondary documents.

Prerequisites – Development SME

Qualifications

It is required that the development SME is both a qualified and trained development SME with expertise in the element under consideration in the context of the tool under whose cybersecurity settings are being reviewed.

Note: The specific qualifications and training of development SMEs necessary to assert expertise in providing accurate information regarding the element under consideration is outside the scope of this document.

Note: The development group is responsible for determining and making available the appropriate SME for the analysis of the element under consideration.

Knowledge

It is required that the development SME have expertise in the element under consideration in the context of the tool under whose cybersecurity settings are being reviewed.

Prerequisites – Security SME

Qualifications

It is required that the threat modeling SME is both a qualified and trained security architect (shown above on title page as **Owner**) as defined by the **NIST NCWF** role SP-ARC-002 and detailed in section **12.7 Security Architect** of the **AVCDL** primary document.

Knowledge

It is required that the cybersecurity SME understands the purpose of cybersecurity-relevant tool settings.

Background Information

It is required that the cybersecurity SME has read and understands the AVCDL **Dynamic Analysis Report**, and **Secure Settings Document** secondary documents. Additionally, that the cybersecurity SME has taken training relevant to this activity.

Prerequisites – DevOps SME

Qualifications

It is required that the DevOps SME is both a qualified and trained DevOps SME with expertise in the build system used for the element under consideration in the context of the tool whose cybersecurity settings are being modified.

Note: The specific qualifications and training of DevOps SMEs necessary to assert expertise in providing accurate information regarding the build system used for the element under consideration is outside the scope of this document.

Note: The DevOps group is responsible for determining and making available the appropriate SME for the modifications to build system needed for the tool under consideration.

Knowledge

It is required that the DevOps SME have expertise in the build system used for the element under consideration in the context of the tool under whose cybersecurity settings need modification.

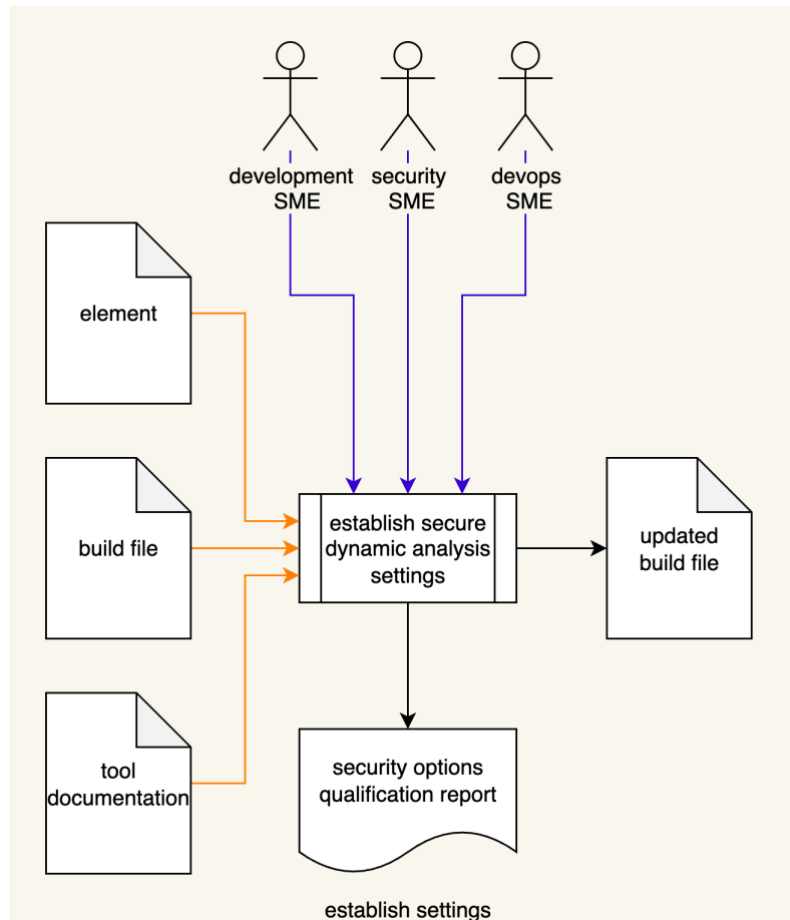
Prerequisites – Input Materials

It is required that the cybersecurity SME acquire all relevant documentation related to the tool under consideration necessary to complete the procedure.

It is required that the development group provide all relevant documentation related to the element under consideration necessary to complete the procedure.

Establish Settings Activity

The workflow diagram for the **establish settings** activity of the **Dynamic Analysis Report** is shown below.

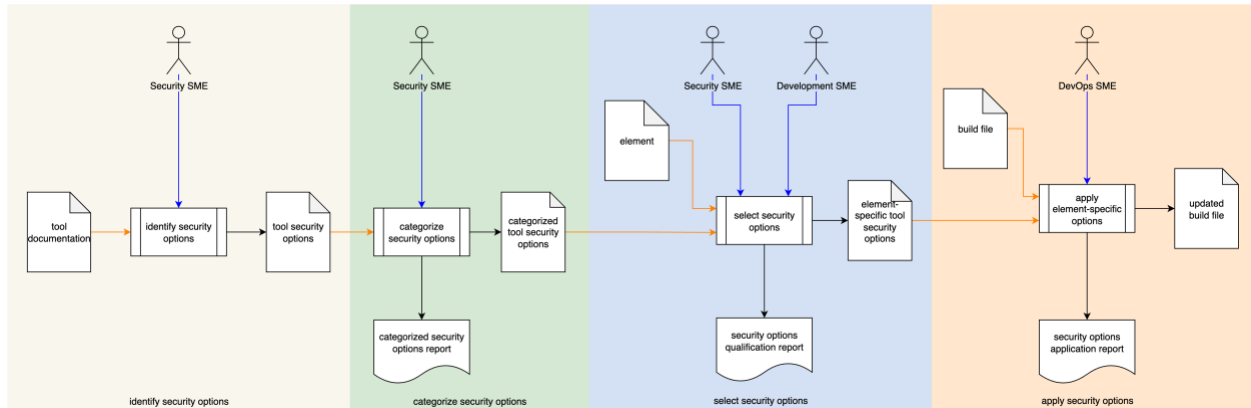


Using the **Element** under consideration, dynamic analysis **Tool Documentation**, and **Build File**; the Development SME, Security SME, and DevOps SME will work together following the **Secure Settings Document** [\[2\]](#) process to determine what element aspect needs to be tested and whether the element will require any kind of instrumentation to enable that testing. An **Updated Build File** will be produced. A **Security Options Qualifications Report** is generated.

Note: The scope of the element may be as small as a single file, or as large as the entire project.

Settings Process Expanded

Consider the following diagram from the AVCDL **Secure Settings Document** secondary document.



Here we can see that the dynamic analysis establish settings activity is more accurately described as a full process in and of itself. It is expressed as a single activity because its steps are common to multiple activities within the **AVCDL**.

Methodology

The methodology to use in this activity is to follow the procedures laid out in the **Secure Settings** procedures [\[4, 5, 6, 7\]](#).

Dynamic Analysis Settings Templates

The dynamic analysis settings may be documented using the **AVCDL tool cybersecurity options template** and **AVCDL element-specific tool cybersecurity options template** Microsoft Excel workbooks [\[8, 9\]](#).

Note: These workbooks are covered in the **AVCDL Secure Settings** tertiary documents [\[4, 5, 6\]](#).

Exit Criteria

This procedure is considered complete once the generated output has been entered into the organization's DMS as a document of record.

Note: The processes and procedures for entering documents into the DMS, or the updating thereof, are outside the scope of this document.

References

1. **AVCDL** (AVCDL primary document)
2. **Dynamic Analysis Report** (AVCDL secondary document)
3. **Secure Settings Document** (AVCDL secondary document)
4. **Secure Settings – Identify Options Procedure** (AVCDL tertiary document)
5. **Secure Settings – Categorize Options Procedure** (AVCDL tertiary document)
6. **Secure Settings – Select Options Procedure** (AVCDL tertiary document)
7. **Secure Settings – Apply Options Procedure** (AVCDL tertiary document)
8. **AVCDL tool cybersecurity options template** (AVCDL template)
9. **AVCDL element-specific tool cybersecurity options template** (AVCDL template)